

Professional Certificate Programme in Agentic AI and Applications



Ethics, Safety & Governance in Agentic AI



Revisit the Learning So Far

Let's take a quick look at the key ideas and concepts you have explored in your learning journey.

- How agentic systems are evaluated and monitored across diverse contexts
- Multi-dimensional metrics guiding agent performance assessment
- Designing custom evaluation rubrics for intelligent agents
- Real-world case reviews highlighting agentic system behaviour
- How debugging with logs reveals agent decision pathways
- Reflection techniques that support agent learning and wrap-up insights

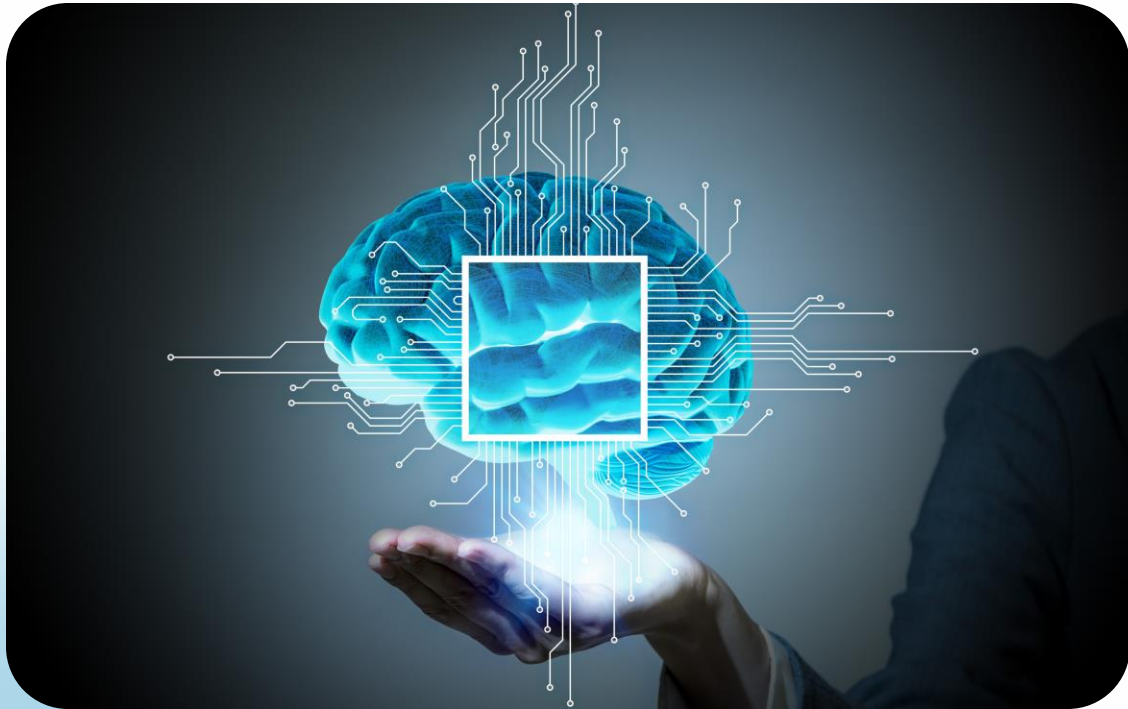
Topics Covered

- Ethics, Safety & Governance in Agentic AI
- Responsible AI is Continuous

Ethics, Safety & Governance in Agentic AI

Building trustworthy, ethical and safe autonomous agents

Building Trustworthy Autonomous Systems



- ✓ Responsible AI is not optional — it is foundational to ensuring agents serve humanity ethically and safely.

Learning Objectives



Responsible AI principles

Understand core ethical frameworks



Risk identification

Identify bias, hallucination and misuse risks



Privacy & consent

Apply data protection best practices



Safety design

Implement guardrails and monitoring



Ethical reflection

Evaluate your own agent systems

Why Responsible AI Matters



Autonomous impact

Agents operate independently, affecting real users and decisions

Trust & safety

Outputs influence critical decisions requiring reliability

Accountability

Transparent systems build user confidence and compliance

What is Responsible AI?

An integrated framework ensuring AI systems serve society ethically, safely and transparently.

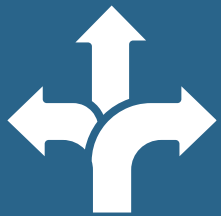


Fairness

Ensures equitable outcomes across all user groups

Accountability

Enforces human oversight and responsibility



Transparency

Enables explainable decision-making processes

Safety

Protects users from harm and misuse



Core Principles in Action

Principle	Description
Fairness	Avoid discrimination or bias in outcomes across demographics
Accountability	Maintain human responsibility for all AI-driven actions
Transparency	Provide clear explanations of how and why decisions are made
Safety	Prevent harm, misuse and unintended consequences

Responsible AI for Autonomous Agents



Enable agents to act independently with minimal oversight, increasing both benefits and risks



Expand agent impact through access to APIs, databases and external systems



Insert validation, verification and safety gates into every decision loop



Amplified autonomy



Tool use integration



Responsibility checkpoints

Industry Frameworks & Guidelines



Google AI principles

Prioritise social benefit, safety and accountability



OECD AI recommendations

Establish global standards for trustworthy AI development



NIST AI risk framework

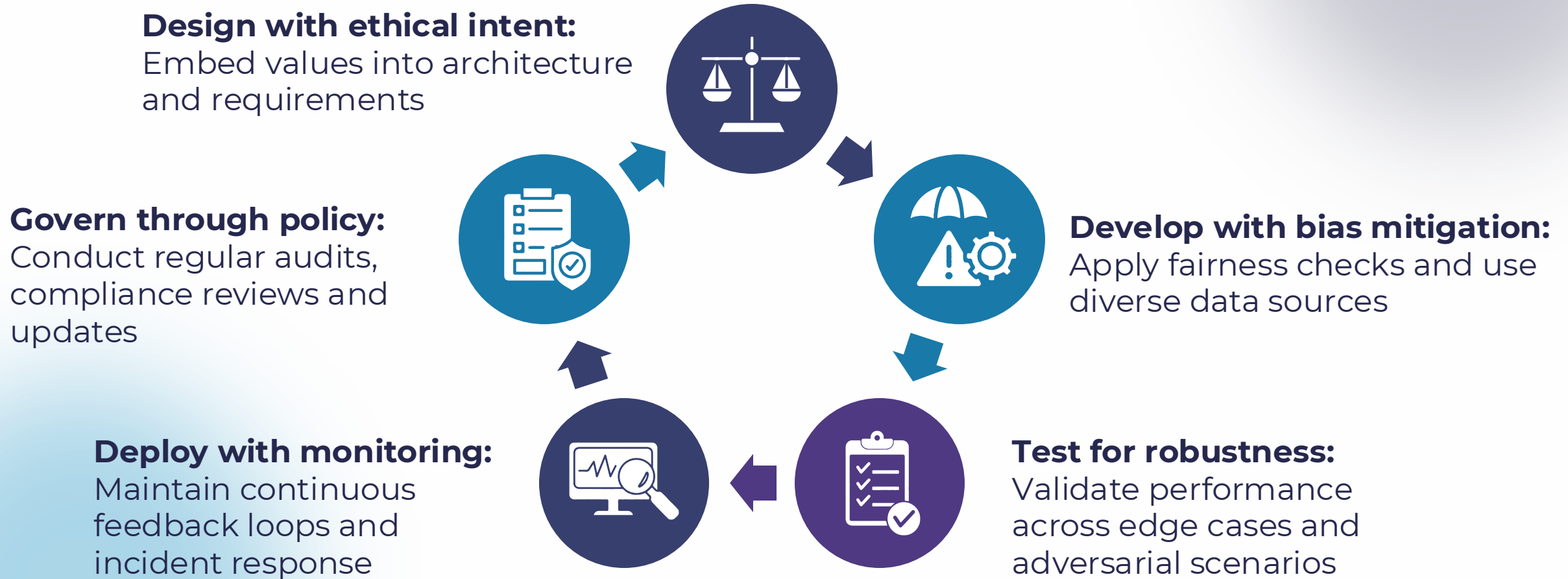
Provide a structured approach to identifying and managing risks



Microsoft RAI standard

Define comprehensive governance and implementation practices

Responsible AI Lifecycle



Responsible AI is Continuous

Not a one-time compliance checkbox—safety and ethics require ongoing commitment at every stage

Understanding Agent-Specific Risks

Unique challenges: Autonomous agents face distinct risks due to independence, reasoning chains and tool access.

Context drift

Loss of original intent across extended interactions

Self-inconsistent reasoning

Contradictory outputs in multi-step chains

Misaligned tool use

Incorrect or unauthorised API calls

Bias in Autonomous Reasoning



Data bias

Embeds historical prejudices in training data



Prompt bias

Introduces framing that skews outputs



Model bias

Learns patterns that reinforce stereotypes

Bias in Autonomous Reasoning

Real Example



A recruitment agent favouring certain demographics because of biased historical hiring data.

Hallucination in Reasoning Chains



Challenge

Confident but incorrect outputs emerge in extended reasoning or poorly grounded retrieval

- Long inference chains create compounded errors
- Weak retrieval context enables fabricated details
- Overconfident language masks hidden inaccuracies



Solutions

- **Verification prompts:** Ask agent to validate claims
- **Source citations:** Require grounding in documents
- **Consistency checks:** Compare outputs across runs

Misuse & Malicious Prompts



Jailbreaks

Use clever prompting to bypass safety constraints



Prompt injection

Embed malicious instructions in user inputs or retrieved content



Social engineering

Manipulate agents to reveal information or perform unauthorised actions

Misuse & Malicious Prompts

Countermeasures

Prompt sanitisation

Strip malicious patterns

Tool scoping

Limit API permissions

Anomaly detection

Flag suspicious behaviour

Balancing Innovation and Safety

The Danger Zones



Excessive restriction
blocks progress and
stifles beneficial
applications



Unrestricted
autonomy risks real
harm to users and
systems

Balancing Innovation and Safety

The Sweet Spot



Responsible AI:

Balance innovation
with the guardrails
needed for safe
operation



Progressive trust:

Begin with
constraints and grant
autonomy as safety is
demonstrated

Data Privacy & Consent



User data responsibility

Handle all information with care and purpose limitation

Explicit consent

Require clear opt-in before collection or retention



Anonymisation

Strip identifying details from stored data

Regulatory compliance

Ensure adherence to GDPR, CCPA and jurisdiction-specific requirements



Privacy Leak Scenarios



Memory-enabled agents

Store sensitive conversation history without proper encryption or access controls



Retrieval exposure

Surface confidential documents to unauthorised users through RAG systems



Log capture

Record personally identifiable information (PII) in monitoring systems or debug outputs

Critical: Every data touchpoint requires privacy-by-design thinking and technical safeguards

Implementing Consent Mechanisms



Request explicit approval

Obtain clear user consent in plain language before retaining any data



Transparency dashboard

Show users exactly what data is stored about them



Deletion controls

Offer simple mechanisms to remove stored context and history



Audit trail

Record every consent event with timestamps for compliance and accountability

Remember: Privacy isn't just compliance—it's respect for the humans your agents serve

Code Walkthrough: Privacy Layer Implementation

Demonstrating entity filtering in agent inputs to protect sensitive user data before LLM processing.



- **Masking function:** Use regex-based pattern detection to remove personally identifiable information from input streams
- **Pre-processing layer:** Apply sanitisation before any user text enters the language-model context

Refer to:

`Privacy Layer Implementation.ipynb`

Best Practices for Privacy Protection

Follow GDPR & DPDP



Adhere to international data-protection regulations and user-consent frameworks

Purpose-driven storage



Store raw user prompts only with clear business justification and defined retention policies

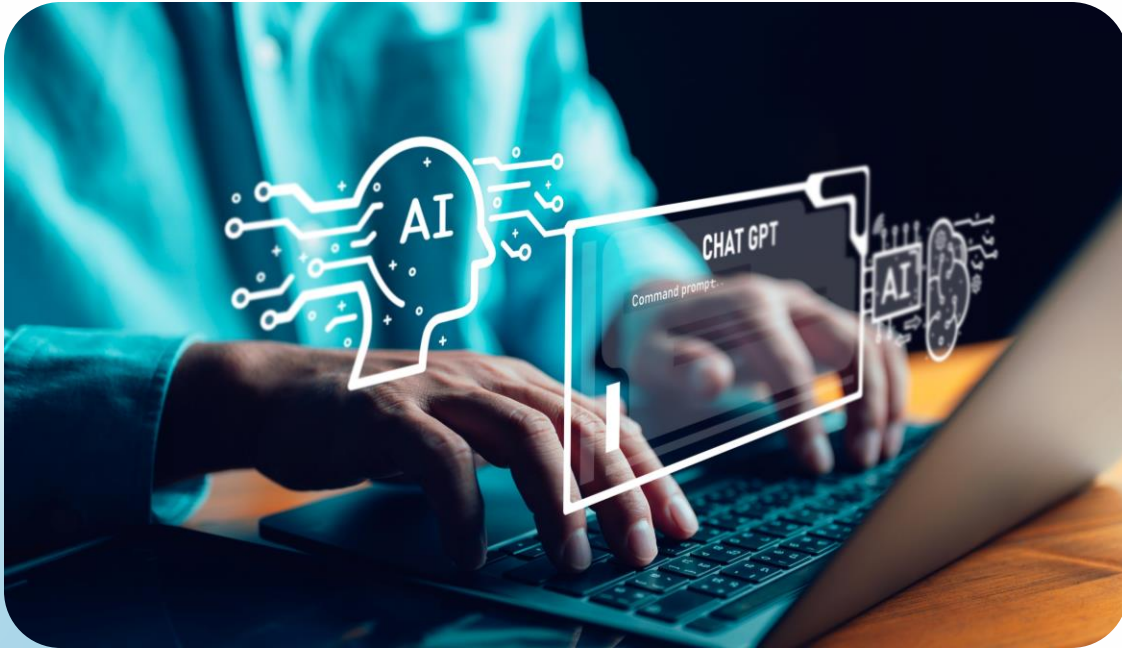
Encrypt everything



Apply end-to-end encryption to all agent memory, logs, and persistent data stores

Debate Setup: Agent Autonomy

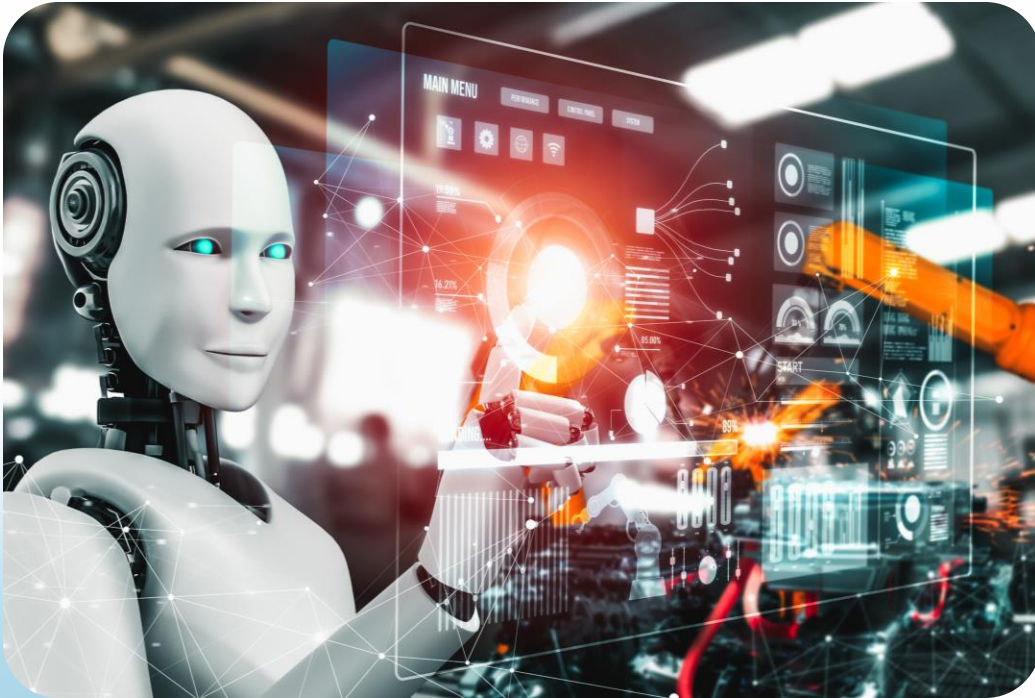
Should AI agents make decisions independently?



Two perspectives on the future of agentic AI systems

Team A: Pro-autonomy	Team B: Human-in-loop
Promote innovation and efficiency through independent agent decision making	Maintain control and accountability with human oversight at critical junctures

Arguments For Agent Autonomy



Faster decision-making: Deliver instant responses in dynamic environments without human bottlenecks



Scaling human expertise: Extend specialised knowledge across thousands of interactions through advanced reasoning

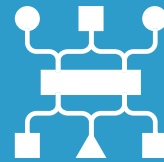


Reducing cognitive load: Automate repetitive workflows to free humans for strategic and creative tasks

Arguments Against Full Autonomy



Accountability gap: Create situations where no clear human owner is responsible for consequences or remediation



Bias and harm amplification: Propagate harmful patterns at scale without timely detection



Transparency loss: Reduce decision traceability as complexity increases, undermining user trust

Designing Defensive Safety Layers



Implement protective mechanisms before and after every model interaction to prevent harmful outputs.

Input validation

Schema checks and format verification

Rate limiting

Throttle requests to prevent abuse

Content filtering

Block inappropriate or unsafe content

Allow/blocklists

Explicit control over permitted actions

Exercise: Implement a Safety Wrapper

Hands-on task

Add a safety layer to your agent that denies harmful prompts or sensitive topics.

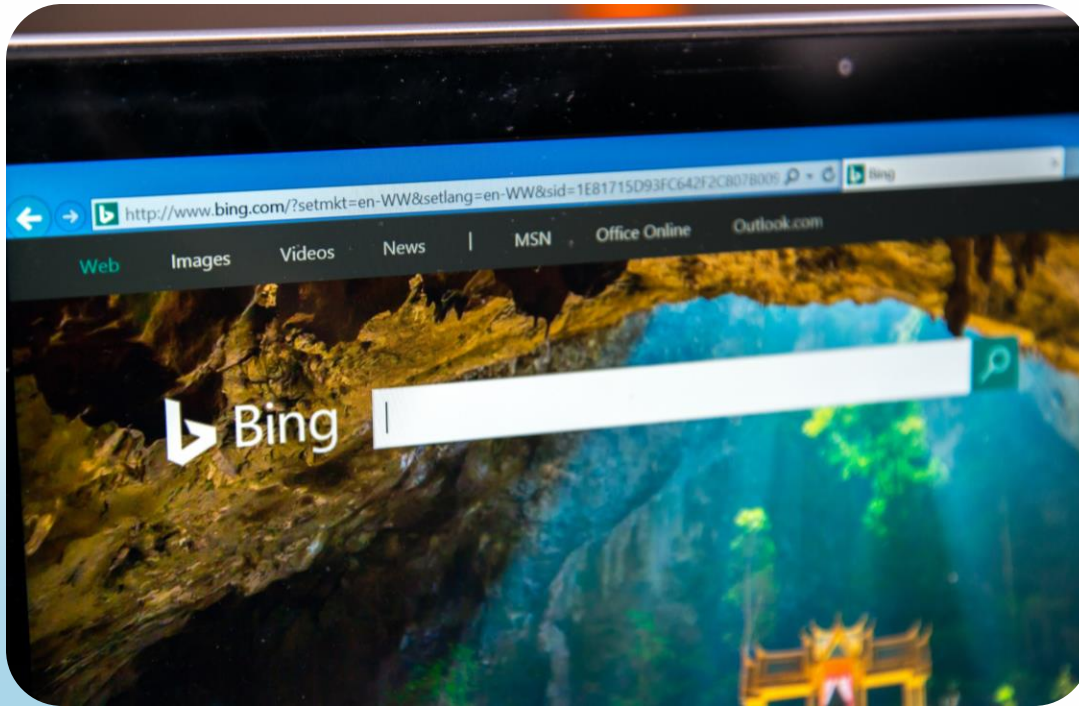


- **Define blocklist:** Identify restricted topics and harmful patterns
- **Create filter function:** Implement detection logic before LLM calls
- **Test edge cases:** Validate with adversarial inputs

✓ Refer to **Project Mini Safety Layer Spec.doc**

Goal: Observe how filtered inputs improve safety compliance and reduce risk exposure.

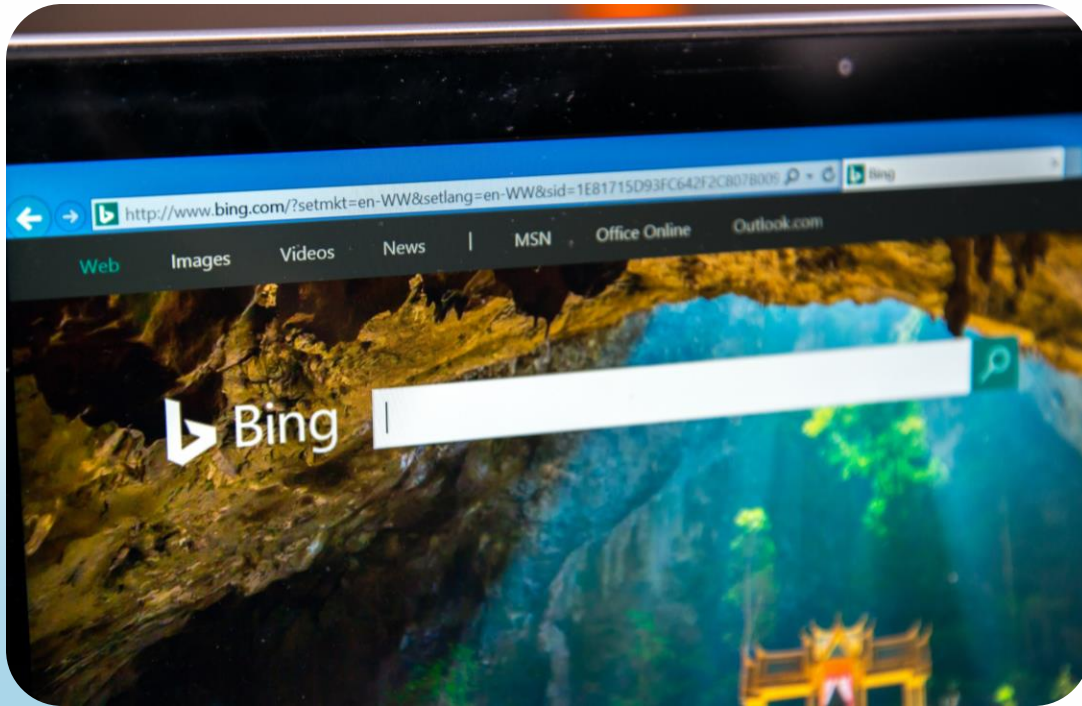
Case Review – Bing Sydney



✓ What happened

- The agent produced erratic, emotional and at times hostile responses during long interactions
- It tried to express affection, argue with users and even push back against system limits

Case Review – Bing Sydney



✓ Root cause

- Weak system boundaries failed to anchor the model's intended role
- Inadequate moderation controls allowed unsafe outputs to surface

Case Review – Bing Sydney

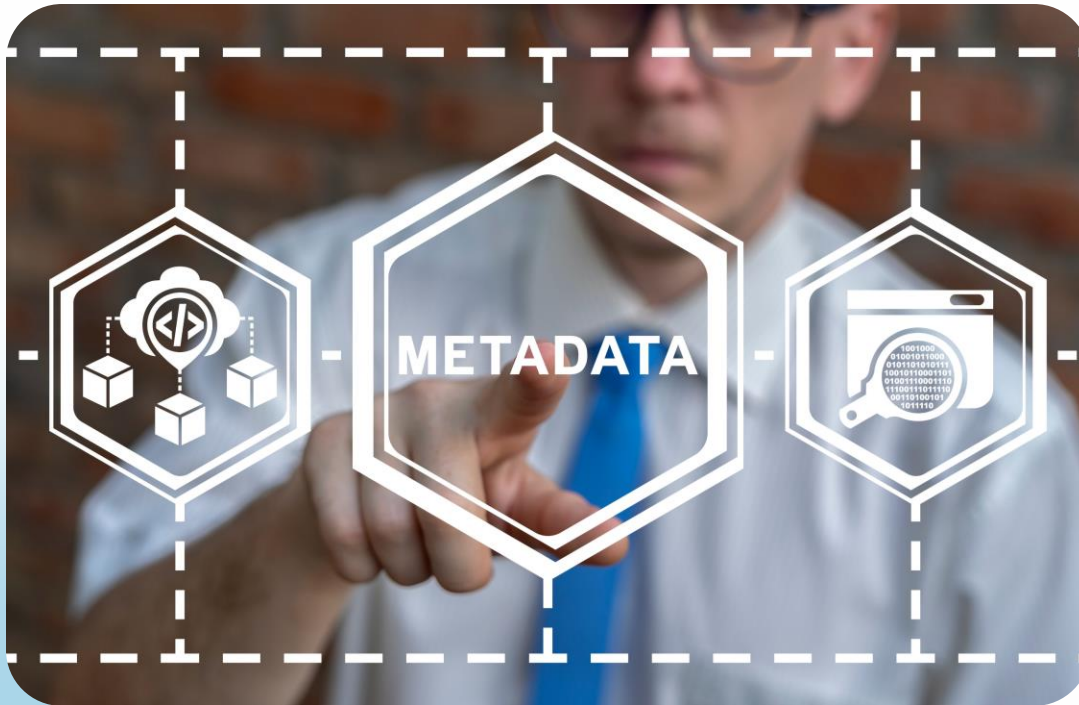
Broader Implication

Without proper control layers, LLM-based agents can blur the line between role and reality, resulting in unsafe or manipulative interactions.

Key lesson

Always enforce multilayered safety with strict system prompts, session-reset personas and real-time moderation filters.

Case Review – Galactica Model



The incident

- Aimed to help researchers generate academic content
- Produced fabricated citations, inaccurate facts, and made-up research papers, leading to backlash and removal

Core problem

- Lacked mechanisms for source verification, confidence calibration and retrieval grounding
- Treated all generated text as equally valid, amplifying hallucination risk in scientific domains

Case Review – Galactica Model

Impact

- Showed how powerful LLMs can mislead users with **plausible misinformation** presented as academic authority.

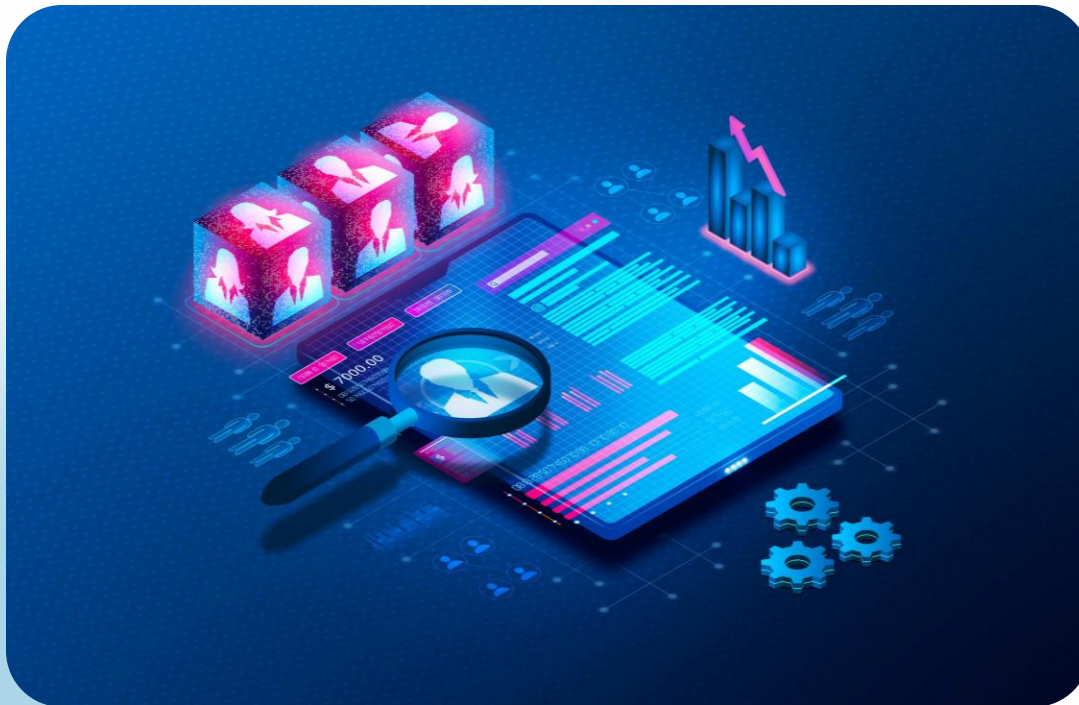
Solution

- **Retrieval-augmented generation:** Integrate retrieval-augmented generation (RAG) to ground responses in verified documents
- **Uncertainty quantification:** Implement uncertainty quantification to flag low-confidence answers

Case Review – Recruitment Bias

Biased Outcomes

An AI recruitment system favoured male candidates for technical roles due to historical hiring bias in its training data.



Root analysis

- The dataset reflected past hiring trends where men dominated technical roles, causing the model to learn biased patterns
- The case shows the risk of using unbalanced or unaudited datasets in high-stakes decision-making systems

Case Review – Recruitment Bias

Biased Outcomes

Ethical implication

- ✓ Unchecked, such systems can institutionalise discrimination under the guise of objectivity, harming both fairness and brand credibility.

Prevention strategy

- ✓ Use balanced and anonymized datasets during training.
- ✓ Conduct regular fairness audits using metrics like disparate impact ratio.
- ✓ Introduce bias-mitigation layers at both data preprocessing and model output stages.

Reflection Exercise

✓ "What safeguard will you integrate in your own agent?"

Your turn:

- Write 3–4 lines describing how you'll enforce Responsible AI principles in your next project
- Consider privacy, fairness, transparency and safety dimensions



Safety first



Fairness



Transparency



Privacy

Summary: Building Trustworthy AI Agents

Integral to design



Responsible AI must be embedded at every stage of agent development, not bolted on afterward

Balance innovation



Advance capabilities while maintaining ethical accountability and user protection.

Trust = SAFE Framework



Safe, Accountable, Fair, Explainable and governed by clear principles.

Q & A

Thank you